MILITARY
SPECIFICATION

MIL-R-55487(EL)
21 APRIL 1966

RECEIVER, RADIO R-1419( )/GR

1. SCOPE

1.1 This specification covers the following units of one type of man-pack or vehicular mounted radio receiver; designated Receiver, Radio R-1419( )/GR:

Cable Assembly, Power, Electrical CX-11606( )/G
Field Pack, Canvas CW-885( )/GR
Mount, Resilient MT-3671( )/GR
Receiver, Radio R-902( )/GR, including:
Canvas Carrying Bag (for R-902( )/GR)
Test Cable Assembly CX-11700( )/G

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied Yarns, Army and Numbered
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Rock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5820

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MIL-B-18	Batteries, Dry
MIL-E-75	Electron Tubes, Packaging, Packing and Container Marking for General Specification For
MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-J-641	Jacks, Telephone, General Specification For
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (For use in Electronic, Communication, and Associated Electrical Equipment)
MIL-L-3891	Luminescent Material and Equipment (Nonradioactive)
MIL-I-8835	Indicator, Humidity, Card, Chemically Impregnated
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment (Supersedes 72-53)
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect (Navy)
MIL-B-43291	Boxes, Fiberboard, Corrugated, Double-Wall, Weather Resistant
MIL-E-55301	Electromagnetic Compatibility

STANDARDS

FEDERAL

FED-STD-751	Stitches, Seams and Stitchings

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MS 3114	Connectors, Receptacle, Electric, Solder Type, Rear Mounting with Nut, Bayonet Coupling (Navy)
MS 3116	Connectors, Plug, Electric, Solder Type, Straight, Bayonet Coupling (Navy)
MS 35179	Connector Receptacle, Electrical, Series BNC, UG-1094A/U
MS 75027	Cable, Cord, Standard Lengths and Standard Tolerances for Communication, Power and Radio Frequency Cable and Cord Assemblies

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester

3. REQUIREMENTS

3.1 Description.- Radio Receiver R-1419( )/GR consists of a single conversion superheterodyne receiver, R-902( )/GR (including canvas carrying case and test cable assembly) capable of receiving AM, FM, CW and MCW signals, in one band, over a frequency range of 95 to 405 mc. The receiver is powered by either an external 24 volts d.c. power source or internal batteries (two BA-405/U (A supply), one BA-415/U (B (1) supply) and one BA-418/U (B (2) supply)). One canvas field pack, for field transportability; a vehicular mount, MX-3671( )/GR; and a power cable assembly CX-11606( )/G are also supplied with each Receiver, R-1419( )/GR.

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3.2 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.3 Preproduction samples.- The contractor shall furnish preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3 and 6.1(e)(1))

3.3.1 Fabrication and assembly.- Preproduction samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The preproduction samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

3.3.2 Preproduction test procedure.- Prior to start of preproduction testing the contractor shall furnish, to the Government, for approval, a complete electrical preproduction test procedure (4 copies). This procedure shall define detail tests for all the requirements of 3.15, 3.16.1 and 3.16.8. The procedure shall include detailed test circuits and essential data on the test equipment, including manufacturer's model, serial number, date of latest calibration, and accuracy. When special test equipment is used, the test procedure will include schematic diagrams and a description of its operation and functions. Sufficient test data to show compliance with the above electrical requirements shall accompany the test procedure. The Government will require 30 days for evaluation of the test data and approval of the test procedure.

3.3.3 Reference standards.- After preproduction samples have been approved, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.4 Parts, materials, and processes; general.- In addition to the requirements of this specification, the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

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3.4.1 Wiring.- All wiring shall be teflon covered.

3.4.2 Fabric and thread.-

3.4.2.1 General.- Fabric and thread shall be preshrunk or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature. (See 4.20)

3.4.2.2 Duck.- Cotton duck shall be in accordance with Federal Specification CCC-C-419, Type I, Number 6 duck.

3.4.2.3 Stitches, seams, and stitching.- Stitches, seams, and stitching shall conform to Federal Standard FED-STD-751.

3.4.3 Cable assemblies.- Length of all electrical cable assemblies shall conform to Standard MS 75027. Power and audio-frequency cable and cord assemblies used for voltages up to and including 600 volts a.c. or corresponding d.c. shall conform to Specification MIL-C-3885.

3.5 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc, and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.2)

3.6 Marking.-

3.6.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4) Front panel marking shall be group I as described in that specification.

3.6.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers.- Radio Receiver R-902( )/GR shall have serial numbers.

3.6.4 Luminescent marking.- Luminescent marking shall be Type P in accordance with Specification MIL-L-3891.

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3.7 Solid-state device protection.- All solid-state circuitry shall be protected from destructive transients originating in the vehicular supply, or any other supply, by the use of semi-conductor devices specifically designed for transient suppression. All solid-state circuits shall be protected against accidental reverse polarity.

3.8 Electrical safety.- Construction shall be such that when the receiver is assembled and set up for operation, personnel shall be protected from contact with potentials in excess of 30 volts to ground, chassis, or frame. Chassis shall not be used as a current carrying conductor.

3.9 Weight and size.- The weight and size of the units of Radio Receiver R-1419( )/GR shall not exceed that of the procurement model.

3.10 Test cable assembly.- Each receiver, R-902( )/GR shall be provided with a 12 inch Test Cable Assembly, CX-11700( )/G. This cable assembly shall be terminated with connectors which will permit operation of the receiver when it is removed from its case. The test cable shall be securely stored in the receiver dust cover.

3.11 Canvas Carrying Case (Receiver R-902( )/GR).- A water repellent, cotton duck carrying case shall be supplied with each Receiver R-902( )/GR. This carrying case shall be similar to the procurement model and shall house the Receiver R-902( )/GR and the Government furnished headset and extension cord.

3.12 Mount, MT-3671( )/GR.- Unless otherwise specified in the bid request and contract a vehicular mount, MT-3671( )/GR shall be furnished with each Receiver R-1419( )/GR. This mount shall hold and protect Receiver R-902( )/GR from being damaged during vehicular transit and operation. Construction of the mount shall be per the procurement model.

3.13 Power Cable Assembly CX-11606( )/G.- The contractor shall supply a 10 foot power cable assembly, CX-11606( )/G, with each receiver R-1419( )/GR. The cable assembly shall be terminated at one end with a MS3116J8-4P connector and other end with a MS3116J8-4S connector in accordance with Specification MIL-C-26482.

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3.14 Field Pack CW-885( )/GR.- A water repellent cotton duck field pack, CW-885( )/GR, shall be supplied with each receiver R-1418( )/GR. This field pack shall be similar to the procurement model and shall be capable of housing the mount MT-3671( )/GR; power cable assembly CX-11606( )/G; and one set of spare batteries (two BA-405/U, one BA-415/U and one BA-418/U).

3.15 Electrical requirements.-

3.15.1 Antenna inputs.- Receiver R-902( )/GR shall be provided with two antenna input receptacles. These antenna inputs shall each have an impedance of 50 ± J100 ohms. A switch shall be provided to select one or the other antenna input. The switch shall afford at least 60 db isolation between inputs. (See 4.7.2 and 4.7.3)

3.15.1.1 Antenna trimmer.- The antenna trimmer shall provide a means by which a signal impressed into the nominal 50-ohm antenna input can be maximized for source impedance ranging from 50 ohms pure resistive to 50 ± J100 ohms. The act of optimizing the antenna trimmer at any frequency within a band shall yield sensitivities at other frequencies within the same band that are within 6 db of the values achievable if the antenna trimmer is retuned at each frequency of measurement.

3.15.2 Frequency range.- The receiver shall cover the frequency range of 95 to 405 mc in one band. (See 4.7.4)

3.15.3 Frequency stability.-

3.15.3.1 Frequency stability vs time.- The stability of the high frequency oscillator and IF center frequency shall be such that the frequency shall not deviate more than the amounts listed when operating from a cold start. (See 4.7.5)

Time (minutes)	Drift (percent)
0 - 5	0.15
5 - 60	0.015

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3.15.3.2 Frequency stability vs temperature.- The frequency drift of the receiver shall not exceed 0.15 percent from -40° F to +150° F. The rate of frequency change shall not exceed ± .0008 percent per degree change in temperature. (See 4.7.6)

3.15.3.3 Frequency stability vs vehicular supply voltage.- The receiver frequency shall not deviate more than ±0.02 percent when the vehicular supply input voltage is raised from 20 to 32 volts d.c. (See 4.7.7)

3.15.3.4 Frequency stability vs battery pack voltage.- The receiver frequency shall not deviate more than ±0.1 percent when the battery pack voltage is varied as listed below. (See 4.7.8)

A supply from 1.5 volts d.c. to 1.4 volts d.c.
B(1) supply from 90 volts d.c. to 72 volts d.c.
B(2) supply from 45 volts d.c. to 38 volts d.c.

3.15.3.5 Frequency stability vs received signal level.- The observed frequency of maximum response, when receiving a 0.25-volt signal at the 50-ohm antenna input, shall not deviate from the frequency observed at the level corresponding to the minimum sensitivity by ± 15 kc. (See 4.7.9)

3.15.4 Sensitivity.-

3.15.4.1 AM and CW sensitivity.- The signal level required to yield a 10 db signal-plus-noise to noise ratio from the receiver audio output shall not exceed the values listed below. (See 4.7.10 and 4.7.11)

Freq. (mc)	AM and CW Sensitivity in uv Dry Batt. Operation	AM and CW Sensitivity in uv Vehicular Operation
95 to 405	3.0	1.0

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3.15.4.2 FM sensitivity.- The signal level required to yield 20 db signal-plus-noise to noise ratio from the receiver audio output shall not exceed the values listed in the following table. A linear interpolation shall serve as the requirement between tabulated frequencies. (See 4.7.12)

Freq (mc)	Sensitivity uv Dry Batt. Operation	Sensitivity in uv Vehicular Operation
100	5.0	2.5
150	5.0	2.0
200	5.0	2.0
300	5.0	2.0
400	5.0	2.0

3.15.4.3 Operating sensitivity vs supply voltage.- The receiver sensitivity shall not deteriorate when the vehicular power supply source voltage drops from 24 to 20 volts or deteriorate more than 6 db when the battery pack voltages are reduced as specified in 3.15.3.4. (See 4.7.13)

3.15.4.4 Operating sensitivity vs temperature.- The operating sensitivity of the receiver at -40° F and +150° F shall not deteriorate more than 6 db from the room temperature requirement. (See 4.7.14)

3.15.5 Frequency calibrator.- The radio receiver shall incorporate a crystal controlled calibration oscillator. Harmonics of the oscillator shall be coupled to the rf portion of the receiver, amplified, hetrodyned with the 1000 cps audio chopper and observed as an audible tone in the output of the receiver. The fundamental frequency of the oscillator shall be 8 mc ±0.005 percent. (See 4.7.15)

3.15.5.1 Calibration oscillator frequency stability and audio output.- The combined frequency stability of the audio chopper and calibration oscillator shall be within ±0.01 percent. At least 10 milliwatts of audio output shall be achieved at any calibration check point.

3.15.5.2 Spurious calibration frequencies.- Spurious calibration frequencies shall be suppressed at least 40 db below the amplitude of the closest check point.

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3.15.5.3 Dial readout error.- Calibration adjustment (See 3.6.16) at any check point shall result in no more than ±2.5 kc or 0.15 percent (whichever is larger) readout error at any point between that check point and adjacent calibration points.

3.15.6 Spurious response.- The receiver shall have an image rejection greater than 60 db during dry battery operation and 80 db during vehicular operation. All other spurious responses, including intermediate frequency rejection, crossmodulation, and intermodulation, shall be suppressed 80 db. (See 4.7.16, 4.7.17, 4.7.18, and 4.7.19)

3.15.7 Radiation and leakage.- The receiver shall not produce more than 400 micromicrowatts of leakage across a dummy load consisting of a 50-ohm termination connected at either antenna receptacle. The receiver shall be so shielded and the oscillator so isolated that the field intensity due to receiver radiation shall not exceed 15 microvolts/meter at 6 feet, at any frequency and for any control settings. In addition, when placed in a field of 250,000 microvolts/meter intensity, the receiver shall have no observable output when the frequency of the field and the receiver tuning are simultaneously varied. The radiation and leakage shall be tested in accordance with the procedures of specification MIL-E-55301.

3.15.8 Dynamic range.- Receiver R-902( )/GR shall not block at any setting of the front panel controls when a 0.5 volt signal, modulated by 1000 cps at 30 percent is impressed at the receiver input. (See 4.7.20)

3.15.9 Intermediate frequency.- The intermediate frequency shall be 30 mc ± 0.5 percent. (See 4.7.21)

3.15.10 IF selectivity.- Two values of IF bandwidth shall be available. Response characteristics shall be symmetrical about the IF center frequency to within ±5 kc and free from irregularities (peak to valley) in excess of 3 db within 60 db of the maximum response. The IF response shall be as listed below. (See 4.7.22 and 4.7.23)

Response	Bandwidth	
	Broad Position	Narrow Position
6 db	160 ± 20 kc	25 ± 3 kc
60 db	Less than 500 kc	Less than 85 kc

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3.15.11 AVC characteristics.- The audio output shall not vary more than 10 db when modulated signal levels from 10 microvolts to 0.25 volt are applied to the 50-ohm antenna input and the receiver is operated in the AM AVC position. (See 4.7.24)

3.15.12 Limiter characteristics.- The audio output shall not vary more than 6 db when modulated signal levels of 10 microvolts to 0.5 volts are applied to the 50-ohm antenna input and the receiver is operated in the maximum gain FM position. (See 4.7.25)

3.15.13 Audio chopper.- The receiver shall contain a diode output level actuated chopper. The diode output level shall be used to activate a 1000 cps audio chopper which is fed to the audio output. The level of diode injection required to activate the audio chopper shall be such that the 1000 cps audio signal will not be detectable in the audio output when the receiver is operated with maximum RF gain in dry battery mode of operation and no signal on the antenna input. (See 4.7.26)

3.15.14 Audio power output.-

3.15.14.1 Vehicular power input voltage.- The audio power output shall be at least 20 milliwatts with a modulated signal input of one microvolt to the 50 ohm antenna input, when the receiver is powered by 20 to 32 volts d.c. (See 4.7.27.1)

3.15.14.2 Low battery operation.- The audio power output shall be at least 10 milliwatts with a maximum signal of one microvolt into the antenna input using 1.4 volts d.c. for the A supply, 72 volts d.c. for the B(1) supply and 38 volts d.c. for the B(2) supply. (See 4.7.27.2)

3.15.15 Audio harmonic distortion.- Audio harmonic distortion shall not exceed 7 percent with 10 milliwatts output. (See 4.7.28)

3.15.16 Audio response.- With 1000 cps as a reference level the overall audio response shall be as follows: (See 4.7.29)

Audio frequency (cps)	Response (db)
250	-5 ± 1
400	-2 ± 2
1000	0

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Audio frequency (cps)	Response (db)
2500	-2 ± 2
3500	-4 ± 2
5000	-10 or greater
7000	-20 or greater

3.15.17 Microphonics.- There shall be no objectionable microphonic noises greater than 3 db above the receiver quiescent value when tested in accordance with 4.7.30. An objectionable microphonic noise is defined as an audible interference, sustained or repetitious, affecting the intelligibility of an incoming signal for a period longer than 3 seconds. (See 4.7.30)

3.15.18 Diode outputs.- Two diode outputs shall be provided; one negative with respect to ground and the other positive with respect to ground. Each output shall provide at least 2.5 volts into an impedance of 50,000 ohms for the signal level specified for minimum AM sensitivity. (See 4.7.31)

3.15.19 Panel meter.- The meter shall indicate signal level, battery voltages and internal power converter outputs. Signal level shall indicate full scale deflection with a 0.005 volt receiver input and less than 10 percent scale with no signal input. The meter shall not be damaged with any input signal level within the dynamic range of the receiver. The meter shall indicate battery conditions with a "GOOD" and "REPLACE" marking. The meter shall indicate a peak reading of at least 25 percent of full scale at all calibration check points. (See 4.7.32)

3.15.20 Internal power converter.- The regulated d.c. to d.c. power converter shall produce outputs of 1.5 ± 0.1 volts d.c., 90^{+2}_{-4} volts d.c. and 45^{+1}_{-2} volts d.c. with an input of 20 to 32 volts d.c. (See 4.7.34)

3.15.20.1 Output voltage ripple.- The output voltage ripple from this power converter shall be 4.0 millivolts rms or less at the 1.5 volt output and 10.0 millivolts rms or less at the 45 and 90 volt outputs. Further, there shall be no ripple and signal intermodulation effects noticeable in the output of the receiver for any proper setting of the receiver controls, when input signal levels within the dynamic range specified herein are employed. Construction shall be predicated on an input line ripple of 0.1 volts rms. (See 4.7.34)

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3.16 Controls and jacks.-

3.16.1 Tuning control and frequency indicator.- The receiver tuning shall be directly calibrated in frequency. The marking interval shall correspond to that of the procurement model. The control shall operate free from backlash and have a resettability of 0.1 percent. (See 4.7.33)

3.16.2 RF gain control/AVC switch.- This control shall be a dual purpose device controlling both RF and IF gain, and the application of AVC. In its extreme counterclockwise position, the RF and IF gain shall be minimum. The gain shall increase with increasing clockwise rotation in logarithmic fashion. In the extreme clockwise position, AVC operation shall be effected when the function switch is set for AM operation. When FM or CW operation is employed, the extreme clockwise position shall yield maximum receiver gain.

3.16.3 AF gain control.- The AF gain control shall produce a near logarithmic variation of the receiver audio gain as the function of rotation.

3.16.4 Meter circuit selector switch.- This switch shall provide a means by which the panel meter may be connected to suitable circuits, to read received signal level and the operating voltages supplied to the receiver when either dry battery or vehicular power is employed.

3.16.5 Antenna trim control.- The antenna trimmer control shall match the receiver input impedance to the antenna impedance.

3.16.6 Dial light switch.- The dial light switch shall operate the 90 milliwatt dial light.

3.16.7 Function switch.- A multiposition detent action switch shall select any of the following modes of operation: AM, FM, CW and Calibrate. In the AM position, the receiver shall be operative for normal AM reception. In the FM position, all necessary circuit changes shall be activated for reception of this emission. In the CW position, the appropriate detector shall be switched in, the BFO energized, and such circuitry changes made as are necessary to obtain efficient CW reception. The Calibrate position shall cause the receiver to function in a manner similar to CW operation; however, the calibration signal shall be injected into the RF circuit.

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3.16.8 Calibration adjustment.- By means of this control, mechanical adjustment of the receiver shall be accomplished whereby the dial-indicated frequency and the actual received frequency shall be made to coincide. The range of this control shall be sufficient to account for all normal variations in receiver parameters. (See 4.7.36)

3.16.9 Power switch.- This switch shall select either dry battery or vehicular power operation of the receiver. The off position shall remove all power from the receiver and the internal power converter. The switch shall have conspicuous "ON" markings when operating on dry battery or vehicular power and "OFF" marking when power has been removed from the receiver circuits.

3.16.10 Bandwidth selector switch.- This switch shall select the appropriate circuits for 25 kc or 160 kc bandwidth.

3.16.11 Front panel connectors.- Front panel connectors shall consist of two BNC type antenna-input connectors, UG-1094 per MS 35179; one audio output jack, type JJ-134 per Specification MIL-J-641/18; one vehicular power input connector MS3114E8-4P and one diode output connector, MS3114E8-3S, per Specification MIL-C-26482.

3.16.12 Rear panel connections.- Duplicate connections shall be mode available at a connector in the dry battery compartment for the audio output, power input, and diode output for use when the receiver is used in a vehicularly mounted mode of operation. The rear panel connector shall be MS3114E10-6P in accordance with Specification MIL-C-26482.

3.17 Dry battery power supply.- The dry batteries listed in 3.1 (per Specification MIL-B-18) shall be capable of being mounted in the battery compartment within the receiver dust cover. This compartment shall have a watertight seal to keep any moisture or battery acid from entering the receiver compartment or leakage out the rear. (See 4.21)

4.18 720 hour run.- When tested as specified in 4.7.35; a failure occuring within the first 72 hours, or if more than 3 failures occur during the remainder of the test, this will be cause for rejection. A failure shall be defined as the need for maintenance other than vacuum tube replacement. The daily tests and the tests at the end of the run shall meet specified requirements.

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3.19 Service conditions.- The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.19.1 Temperature.- (See 4.8)

(a) Operating: Ambient temperature in the range of +150° F to -40° F. (The 150° F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160° F to -65° F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.19.2 Relative humidity.- (See 4.9) Up to 97 percent relative humdity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.19.3 Elevation (See 4.10).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.19.4 Rainfall (See 4.11).- Two inches of rainfall per hour with wind at 20 miles per hour.

3.19.5 Sand and dust (See 4.12).- As encountered during heavy traffic over dry clay roads or in desert areas.

3.19.6 Vibration, bounce and shock.-

3.19.6.1 Vibration, internal.- With the equipment shock-mounts blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.13)

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3.19.6.2 Vibration, equipment.- The natural frequency of the equipment, mounted on its shock-mounts, shall be between 25 and 35 cycles per second. The amplitude of vibration of the mounted equipment shall not exceed three times the amplitude of the applied vibration at any frequency between 10 and 25, or between 35 and 55, cycles per second. (See 4.14)

3.19.6.3 Bounce and shock.- The equipment shall be capable of meeting the requirements of Table I.

TABLE I - BOUNCE AND SHOCK

Paragraph Heading	Paragraph Number	Performance after test
Bounce; vehicular mounted	4.15	Specified performance. (Note A) No physical damage.
Bounce; field transportability	4.16	Specified performance. (Note A) No physical damage except minor surface abrasion.
Shock; bench-handling	4.17	Specified performance. (Note A) No physical damage.
Shock; drop	4.18	Specified performance. (Note A) Any physical damage shall be minor only.

Note A: The equipment shall meet specified performance for the following:
AM, CW and FM sensitivity 3.15.4
Audio chopper 3.15.13
Audio output 3.15.14
Diode output 3.15.18

3.19.7 Orientation (See 4.19).- Any orientation up to 20 degrees from normal operating position (that is forward, backward, left or right).

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3.20 Electrical burn-in time.- After final alignment and prior to preliminary electrical inspection each receiver R-902( )/GR on order shall be operated, using the standard vehicular power input, for 50 hours. The contractor shall make all necessary repairs and realignments to correct any receiver failures during and at the completion of this burn-in time.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.23) Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Electromagnetic compatibility.- The equipment shall meet the emanation and susceptibility limits of Specification MIL-E-55301 for tactical equipment (See 4.7.37). Application of suppression components and techniques shall conform to that specification. The regulated vehicular power supply shall, as a separate item, meet the requirements of Specification MIL-E-55301.

3.23 Adjustment and repair.- The equipment shall be so constructed that parts, terminals, wiring, etc, are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring and with use of the minimum number and variety of special tools, particularly those needed for tuning and adjustment. The contractor shall furnish all special tools required, with each receiver.

3.24 Final operational inspection.- Immediately prior to preparation for delivery receiver R-902( )/GR shall be given a final operational inspection as specified in 4.22, to determine that the receiver is operable.

3.25 Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (Also see 4.5).

3.26 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.1).

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3.27 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

In Specification MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Flux and cleaning agents for soldering.
Process for soldering.
Cleaning.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process for welding.
Mechanical operation of controls.
Gears.
Self-tapping screws.
Fasteners for access devices.
Fabric and thread.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

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(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.25)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table II, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables III, IV, and V, respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, bench-handling, (4) shock, drop, and (5) rainfall; other preproduction inspections may precede, follow, or be interspersed between the foregoing.

TABLE II - PREPRODUCTION INSPECTION

Inspection (For Additional preproduction inspection see 4.3)	Requirement Paragraph	Inspection Paragraph
Sand and dust test	3.19.5	4.12
Test for internal vibration	3.19.6.1	4.13
Shock test; bench-handling	3.19.6.3	4.17
Shock test; drop	3.19.6.3	4.18
Oreintation test	3.19.7	4.19

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.8) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to Group A, Group B, or Group C inspection, except preproduction samples, shall be preconditioned after final assembly. (See 3.25)

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table III and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.22) shall be last. (See 6.5)

TABLE III - GROUP A INSPECTION

Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical				
Cable Assembly CX-11606( )/G	3.27	4.24	1%	4%

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TABLE III - GROUP A INSPECTION (Continued)

Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Cable Assembly CX-11700( )/G	3.27	4.24	1%	4%
Canvas Carrying Bag (for R-902( )/GR)	3.27	4.24	1%	4%
Field Pack CW-885( )/GR	3.27	4.24	1%	4%
Mount MT-3671( )/GR	3.27	4.24	1%	4%
Receiver, R-902( )/GR	3.27	4.24	1.5 dphu	6.5 dphu
Electrical			1.5 dphu for the group	All electrical defects are considered major
Frequency range	3.15.2	4.7.4		
Frequency stability vs vehicular supply voltage	3.15.3.3	4.7.7		
Frequency stability vs low battery voltage	3.15.3.4	4.7.8		
Frequency stability vs received signal level	3.15.3.5	4.7.9		
AM sensitivity	3.15.4.1	4.7.10		
CW sensitivity	3.15.4.1	4.7.11		
FM sensitivity	3.15.4.2	4.7.12		
Sensitivity vs supply voltage	3.15.4.3	4.7.13		

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TABEL III - GROUP A INSPECTION (Continued)

Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Frequency calibrator	3.15.5	4.7.15		
Spurious calibration frequencies	3.15.5.2	4.7.15		
Calibration oscillator frequency stability and audio output	3.15.5.1	4.7.15		
Dial readout error	3.15.5.3	4.7.15		
Dynamic range	3.15.8	4.7.20		
Intermediate frequency	3.15.9	4.7.21		
AVC characteristics	3.15.11	4.7.24		
Limiter characteristics	3.15.12	4.7.25		
Audio chopper	3.15.13	4.7.26		
Audio output (vehicular)	3.15.14.1	4.7.27.1		
Audio output (low battery)	3.15.14.2	4.7.27.2		
Audio harmonic distortion	3.15.15	4.7.28		
Audio response	3.15.16	4.7.29		
Microphonics	3.15.17	4.7.30		
Diode outputs	3.15.18	4.7.31		

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TABLE III - GROUP A INSPECTION (Continued)

Inspection	Rqt Para	Insp Para	AQL Major	AQL Minor
Panel meter	3.15.19	4.7.32		
Tuning control and frequency indicator	3.16.1	4.7.33		
Calibration adjustment	3.16.8	4.7.36		
Power converter	3.15.20	4.7.34		
Operational	3.24	4.22		

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table IV and to the special procedures for small-sample inspection of Standard MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in TABLE IV, shall be as follows:

AQL	Inspection level for normal, reduced and tightened inspection
4.0%	S-3
6.5%	S-3

4.5.2.2 Order of inspection within Group B.- The inspection shall be performed in any order which is satisfactory to the Government.

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TABLE IV - GROUP B INSPECTION

Inspection	Rqt Para	Insp Para	AQL (See 4.5.2.1)
50 ohm antenna input impedance	3.15.1	4.7.2	4.0%
Antenna selector switch	3.15.1	4.7.3	4.0%
Frequency stability vs time	3.15.3.1	4.7.5	4.0%
Image rejection	3.15.6	4.7.16	4.0%
Intermediate frequency rejection	3.15.6	4.7.17	4.0%
Intermodulation distortion products	3.15.6	4.7.18	4.0%
Crossmodulation	3.15.6	4.7.19	4.0%
Intermediate frequency selectivity	3.15.10	4.7.22;4.7.23	4.0%
Inspection for dimensional interchangeability	3.21	4.7.23	6.5%

4.5.3 Group C inspection.- This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

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TABLE V - GROUP C INSPECTION

Inspection	Requirement Para	Inspection Para
Frequency stability vs temperature	3.15.3.2	4.7.6
Sensitivity vs temperature	3.15.4.4	4.7.14
720 hour test	3.18	4.7.35
Temperature test	3.19.1	4.8
Moisture resistance test	3.19.2	4.9
Elevation test	3.19.3	4.10
Rain test	3.19.4	4.11
Equipment vibration test	3.19.6.2	4.14
Vehicular bounce test	3.19.6.3	4.15
Field transportability bounce test	3.19.6.3	4.16
Shrinkage test	3.4.2.1	4.20
Battery compartment watertight test	3.17	4.21
Electromagnetic compatibility	3.22	4.7.37

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4.5.3.1 Sampling for inspection of equipment.- Two items of production (excluding preproduction samples) for each Group C inspection shall be selected from the first 50 items produced and from each 100 items produced thereafter. The items shall be selected without regard to their quality and shall be selected from the first units produced of each group.

4.5.3.2 Noncompliance.- If a sample unit fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4)

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Bounce preconditioning.- The unit, with shock mounts removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Electrical tests.-

4.7.1 Standard test conditions.- Unless otherwise specified the electrical tests shall be conducted under the following conditions:

(a) Temperature and humidity----------room ambient

(b) Input Voltage

Vehicular supply----------24 ± 0.5 volt d.c.

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Dry battery

"A" supply ----------1.5 ± 0.1 volt d.c.
"B(1)" supply ----------90 ± 2.0 volts d.c.
"B(2)" supply----------45 ± 1.0 volt d.c.

(c) RF input-----signal generator with 50 ohm output impedance using a 20 db pad in series

(d) Bandwidth

AM--------25 kc
FM-------160 kc

(e) Modulation

AM---------30 percent at 1000 cps
FM---------±30 kc deviation at 1000 cps

(f) Audio output load impedance-------600 ohms

4.7.2 50 ohm antenna input impedance.- Measure the impedance of the 50 ohm antenna input at the low, center and high frequency and 3 equally spaced frequencies in between, with a suitable impedance bridge, to show compliance with 3.15.1 and 3.15.1.1.

4.7.3 Antenna selector switch.- Apply a one microvolt AM modulated signal to one antenna input. Adjust the antenna trim, RF gain and AF gain for 10 milliwatts output. Change the antenna selector switch to the other antenna input position. Increase the input signal until 10 milliwatts output is obtained. The isolation between the two antenna inputs shall be as specified in 3.15.1.

4.7.4 Frequency range.- Each receiver band shall be tested, with a frequency standard, to determine compliance with 3.15.2. The band shall be completely scanned with a sweep generator to ascertain that there are no dead spot within the receiver frequency range.

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4.7.5 Frequency stability vs time.- Operate the receiver in the AM mode at 155 mc and apply a one microvolt modulated signal to the antenna input. Measure the receiver RF frequency drift, with a frequency counter for compliance with 3.15.3.1.

4.7.6 Frequency stability vs temperature.- Subject the receiver to +150° F for 4 hours or until thermal equilibrium is attained (whichever occurs first). Turn on receiver and tune to 155 mc. Allow the receiver oscillator to stabilize. Apply an AM modulated RF signal to the 50 ohm antenna input and monitor with a frequency counter. Lower the temperature to -40° F in 4 steps, 100° F, 50° F, 0° F, and -40° F. Hold the temperature at each step until thermal equilibrium is attained. Measure the frequency drift at each step for compliance with 3.15.3.2.

4.7.7 Frequency stability vs vehicular supply voltage.- The test frequency shall be 155 mc. Apparatus shall be provided to rapidly change the vehicular supply voltage from 20 to 32 volts d.c. The receiver shall be tuned to a stable AM, one microvolt, 155 mc signal. The receiver shall be allowed to stabilize with an input of 20 volts d.c. Change in receiver RF frequency shall be measured as the input supply is abruptly changed to 32 volts d.c. (See 3.15.3.3)

4.7.8 Frequency stability vs battery pack voltage.- With the receiver operating in the internal battery mode, apply 1.5 volts d.c. to the "A" battery supply input, 90 volts d.c. to the "B(1)" supply input and 45 volts d.c. to the "B(2)" battery supply input. Tune the receiver to 155 mc. Apply a one microvolt, AM signal to the 50 ohm antenna input and adjust the frequency for maximum receiver output. Allow receiver frequency to stabilize. Reduce the input voltages as specified in 3.15.3.4. After one hour of operation at this reduced voltage, retune the input signal for maximum receiver output (without changing receiver controls). The frequency difference shall not exceed that specified in 3.15.3.4.

4.7.9 Frequency stability vs received signal level.- Apply a 1 microvolt AM 155 mc test signal to the 50 ohm antenna input and monitor it with a frequency counter. Tune the receiver to the test signal and adjust output for 10 milliwatts. Increase the monitored test signal to 0.25 volt and retune it for maximum receiver output (without changing receiver frequency control). (See 3.15.3.5)

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4.7.10 AM sensitivity.-

4.7.10.1 Vehicular operation.- Operate the receiver in the vehicular, AM mode. Apply a modulated RF signal to the 50 ohm antenna input. The minimum signal level required to produce a 10 db signal-plus-noise to noise ratio shall be determined at the high, center, and low frequencies and 5 equally spaced frequencies in between. The sensitivity figures shall be as specified in 3.15.4.1.

4.7.10.2 Dry battery operation.- Operate the receiver in the internal (dry) battery mode and perform tests in 4.7.10.1. (See 3.15.4.1)

4.7.11 CW sensitivity.-

4.7.11.1 Vehicular operation.- Operate the receiver in the vehicular, CW mode. Apply an unmodulated RF signal to the 50 ohm antenna input. Tune the input signal to a maximum receiver output. Adjust the signal input, AF and RF gain until a 10 db signal-plus-noise to noise ratio is obtained. This test shall be performed on the high, center, and low frequencies and 5 equally spaced frequencies in between. The sensitivity figures shall be as specified in 3.15.4.1.

4.7.11.2 Dry battery operation.- Operate the receiver in the internal (dry) battery mode and perform the tests in 4.7.11.1. (See 3.15.4.1)

4.7.12 FM sensitivity.-

4.7.12.1 Vehicular operation.- Operate the receiver in the vehicular, FM mode. Apply a modulated RF signal to the antenna input. The minimum signal level required to produce a 20 db signal-plus-noise to noise ratio shall be determined at the high, center, and low frequencies and 5 equally spaced frequencies in between. The sensitivity figures shall be as specified in 3.15.4.2.

4.7.12.2 Dry battery operation.- Operate the receiver in the internal (dry) battery mode and perform the tests in 4.7.12.1. (See 3.15.4.2)

4.7.13 Operating sensitivity vs supply voltage.- Perform the sensitivity tests of 4.7.10, 4.7.11 and 4.7.12 under the reduced operating voltages specified in 3.15.3.4. The sensitivities shall meet the requirements of 3.15.4.3.

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4.7.14 Operating sensitivity vs temperature.- Perform the sensitivity tests of 4.7.10, 4.7.11 and 4.7.12 at the temperatures specified in 3.15.4.4. The sensitivities shall meet the requirement of 3.15.4.4.

4.7.15 Frequency calibrator.- Receiver R-902( )/GR shall be tested for compliance with 3.15.5, 3.15.5.1, 3.15.5.2 and 3.15.5.3. The tests shall be performed at each calibration check point.

4.7.16 Image rejection.- Apply a 10 microvolt AM modulated signal to the receiver 50 ohm antenna input. Adjust receiver gain controls for 10 milliwatts receiver output. Without changing the receiver controls, tune the signal generator to the receiver image, and increase the signal input until a 10 milliwatt output is again obtained. The image frequency rejection ratio shall be as specified in 3.15.6 and the test made at the high, middle and low frequencies of the band.

4.7.17 IF rejection.- The IF rejection shall be tested using the procedure of 4.7.16, except the signal generator shall be retuned to the intermediate frequency instead of the image frequency. (See 3.15.6)

4.7.18 Intermodulation distortion products.- The receiver operating in the vehicular mode shall be tested for intermodulation distortion products as specified in 3.15.6.

4.7.19 Crossmodulation.- The receiver, operating in the vehicular mode, shall be tested for crossmodulation as specified in 3.15.6.

4.7.20 Dynamic range.- Operate the receiver at 155 mc, in the AM vehicular mode and apply a one microvolt modulated signal to the 50 ohm antenna input. Adjust the receiver controls for maximum receiver output (no AVC). Slowly increase the input signal to 0.25 volt. The 1000 cps modulation shall be discernable from one microvolt to 0.25 volt. (See 3.15.8)

4.7.21 Intermediate frequency.- Apply a modulated signal to the IF input and adjust input frequency for maximum response. Measure the input frequency with a frequency counter for compliance with 3.15.9.

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4.7.22 Intermediate frequency selectivity 6 db.- Feed at 2.5 microvolt signal to the 50 ohm antenna input. Adjust the receiver controls for -1 volt diode output. Increase the input signal level by 6 db. Increase input frequency until a -1 volt diode output is again obtained. Measure this frequency. Decrease the input frequency until a -1 volt diode output is obtained. Measure this frequency. The bandwidth, symmetry, and irregularities shall be as specified in 3.15.10. Perform this test in both the 25 and 160 kc bandwidth positions.

4.7.23 Intermediate frequency selectivity, 60 db.- Perform the test in 4.7.22 using 60 db increase in signal level instead of 6 db. (See 3.15.10)

4.7.24 AVC characteristics.- Operate the receiver in the AM mode with the AVC on. Apply a 10 microvolt modulated signal to the 50 ohm antenna input and adjust the audio gain control for 10 milliwatts audio output. Increase the input signal level to 0.25V. The audio output shall be as specified in 3.15.11.

4.7.25 Limiter characteristics.- Apply a 10 microvolt FM signal to the receiver antenna input and adjust the receiver for maximum audio output. Increase the receiver input to 0.5 volt. The audio output shall not vary more than that specified in 3.15.12.

4.7.26 Audio chopper.- Operate the receiver in the battery CW mode and adjust for maximum audio output. Slowly tune the receiver through its frequency range while monitoring the audio output for a 1000 cps chopper signal. (See 3.15.13)

4.7.27 Audio power output.-

4.7.27.1 Vehicular power input voltage.- Operate the receiver in the vehicular AM mode with a supply voltage of 32 volts d.c. Apply a modulated signal to the 50 ohm antenna input and adjust the receiver for maximum audio output. The audio output shall be as specified in 3.15.14.1. This test shall be performed at the frequency of lowest sensitivity. Repeat the above test with a supply voltage of 20 volts d.c.

4.7.27.2 Low battery voltage.- Operate the receiver in the internal (dry) battery AM mode with the low A and B supply voltages specified in 3.15.14.2. Apply a modulated signal to the 50 ohm antenna input, and with the receiver controls adjusted for maximum gain the audio output shall be no less than that specified in 3.15.14.2. This test shall be performed at the frequency of lowest sensitivity.

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4.7.28 Audio harmonic distortion.- Apply a 10 microvolt modulated signal to the 50 ohm antenna input. Adjust the audio output for 10 milliwatts. Measure the distortion of the audio output. (See 3.15.15)

4.7.29 Audio response.- Operate the receiver in the AM mode and apply a 1000 cps, 30 percent AM modulated RF signal to the 50 ohm antenna input. Adjust the receiver for 10 milliwatts audio output. Vary the audio modulation to the frequencies specified in 3.15.16. Maintain 30 percent modulation for all audio frequencies. The audio response shall comply with 3.15.16.

4.7.30 Microphonics.- Operate the receiver in the vehicular AM mode, with 32 volts d.c. vehicular supply voltage. Feed a 10 microvolt modulated signal to the 50 ohm antenna input and adjust the receiver for maximum audio output. Remove the modulation and tap the front panel, top, and both sides of the receiver with a rubber mallet having a kinetic energy content of 4 ft/lb. This test shall be performed at the frequency of greatest sensitivity. (See 3.15.17)

4.7.31 Diode outputs.- Apply a one microvolt modulated signal to the 50 ohm antenna input, and with the receiver operating in the AM mode, adjust the controls for maximum audio output. Measure the diode outputs for conformance to 3.15.18. This test shall be performed at the high, center and low frequencies of the band.

4.7.32 Panel meter.- The panel meter shall be tested for each of the functions specified in 3.15.19, in each mode of receiver operation.

4.7.33 Tuning control and frequency indicator.- Operate the receiver in the CW mode and apply a 2.5 microvolt signal to the 50 ohm antenna input. Tune the receiver frequency control for maximum audio output. Note the dial frequency and vernier scale reading. Tune the receiver frequency control 1/2 revolution clockwise, then slowly return to the maximum output. Note the dial frequency and vernier scale reading. Repeat the above procedure turning the frequency control counterclockwise. The dial resettability shall be as specified in 3.16.1.

4.7.34 Power converter.- Apply 20 volts d.c., with 0.1 volt rms ripple voltage, to the input of the d.c. to d.c. converter. The output circuits shall be loaded to simulate receiver R-902( )/GR. Measure the output voltages and output voltage ripple. Increase the input voltage to 32 volts d.c. and maintain the input ripple at 0.1 volt rms. Under bothe conditions the output voltage and ripple shall be as specified in 3.15.20 and 3.15.20.1.

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4.7.35 720 hour test.- The radio set shall be operated continuously for a period of 720 hours. The frequency of lowest sensitivity (4.7.10) shall be used. At the start of the test, the AM sensitivity at the specified frequencies of (4.7.10), frequency calibrator (4.7.15), audio chopper (4.7.26), audio power output (4.7.27) and diode outputs (4.7.31) shall be checked and the results recorded. At the end of each 24 hour period the tests above with the exception of the diode calibration accuracy test, shall be repeated at the frequency of lowest sensitivity only. At the end of the 720-hour run, all of the above tests shall be made at the specified frequencies. The receiver performance shall comply with 3.18.

4.7.36 Calibration adjustment.- Operate the receiver in the calibration mode. Tune the receiver to each calibration point and adjust the calibration adjustment control above and below the calibration point. (See 3.16.8)

4.7.37 Electromagnetic compatibility.- Subject receiver R-902( )/GR and the regulated power converter to the applicable electromagnetic compatibility tests of Specification MIL-E-55301. (See 3.22)

4.8 Temperature test.- The equipment (receiver, cable assemblies and canvas bags) shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

Step	Paragraph	Measurement	Performance
1	(1) 4.7.7	Frequency stability vs Vehicular supply voltage	Full Specification
	(2) 4.7.10.1	AM sensitivity (Vehicular operation)	Full Specification
	(3) 4.7.11.1	CW sensitivity (Vehicular operation)	Full Specification
	(4) 4.7.15	Calibration accuracy	Full Specification
	(5) 4.7.26	Audio chopper	Full Specification
	(6) 4.7.27.1	Audio power output	Full Specification
	(7) 4.7.31	Diode outputs	Full Specification

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Step	Paragraph	Measurement	Performance
3		(1) through (7) of Step 1	Full Specification
5		(1) through (7) of Step 1	Full Specification
8	(1)	(1),(3),(4),(5),(6),(7) of Step 1	Full Specification
	(2) 4.7.14	Operating sensitivity vs temperature	Full Specification
10		(1) through (7) of Step 1	Full Specification

NOTE: At step 8 all panel controls, connectors and cable assemblies shall be operated to ascertain proper function. All canvas items shall be inspected for their service usage.

4.9 Moisture-resistance test.- The equipment shall be tesred as follows:

4.9.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Place all items on contract, that are listed in 1.1, in the test chamber in a manner simulating service usage, except receiver R-902( )/GR chassis shall be removed from its dust cover.

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4.9.2 Test method.-

(a) Dry at 130° ±5° F for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 60 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.9.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The receiver R-902( )/GR shall be operated during the period shown on the standard (no measurements taken).

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.9.3.

4.9.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance
4.7.7	Frequency stability vs Vehicular supply voltage	Full Specification Performance
4.7.10.1	AM sensitivity (Vehicular Operation)	Full Specification Performance
4.7.11.1	CW sensitivity (Vehicular Operation)	Full Specification Performance

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Paragraph	Measurement	Performance
4.7.15	Calibration accuracy	Full Specification Performance
4.7.26	Audio chopper	Full Specification Performance
4.7.27.1	Audio power output	Full Specification Performance
4.7.31	Diode outputs	Full Specification Performance

NOTE: At the completion of this test, Group A electrical tests of Specification MIL-C-3885 shall be performed on all cable assemblies. All panel controls and connectors shall be operated to ascertain proper function. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding of rotating parts. All canvas items shall be inspected for deterioration (seams, zippers and moisture fungus proofing treatment).

4.9.4 Failure.- If the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.9.2(e), it does not pass the test.

4.10 Elevation test.- The receiver, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment).

(a) With the equipment in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.7.10.1	AM sensitivity (vehicular operation)
4.7.31	Diode outputs

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(b) Reduce the barometric pressure to 20.6 ± 0.1 inch of mercury (simulated 10,000 feet above sea level) and, after temperature pressure stabilization, repeat the measurements specified in 4.10(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inch of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.10(a). The equipment shall meet specified performance for these measurements.

4.11 Rain test.- The equipment shall be tested as follows:

4.11.1 Dry at 150° ± 5° F for 48 hours.

4.11.2 Condition for 4 hours at 77° ±5° F and 40 to 60 percent relative humidity.

4.11.3 Take the measurements listed in 4.9.3 (Moisture resistance) to insure that the equipment meet specified performance.

4.11.4 Place the equipment in its normal operating position, in a test chamber maintained at 77° ±5° F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5° F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.11.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for 1 hour.

4.11.6 Rotate the equipment 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for 1 hour.

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4.11.7 Continue as in 4.11.6, until all four sides of the equipment have been exposed to the wind and rain, each side for 1 hour.

4.11.8 Immediately after completion of the foregoing, repeat the measurements specified in 4.9.3 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The equipment shall meet specified performance.

4.11.9 Open the equipment. There shall be no visible evidence of leakage.

4.12 Sand and dust test.- The equipment shall be tested as follows:

(a) Place the equipment in its normal operating position, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, U.S. Standard Sieve Series	Percent
100-mesh	100
140-mesh	98 ± 2
200-mesh	90 ± 2
325-mesh	75 ± 2

(c) Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition losses	0 to 2

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(d) Maintain internal temperature of test chamber at 77° ±4° for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After 6 hours at above conditions, raise the temperature to 160° ±4° F and maintain it at this value for 6 hours. The other conditions shall remain the same.

(f) Remove the equipment from test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the equipment. However, dust shall not be removed by either blast or vacuum cleaning. The equipment shall meet specified performance for the measurements listed in 4.9.3.

(g) Open the equipment. There shall be no evidence of deterioration of either external or internal parts.

4.13 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.19.6.1:

(a) Secure the equipment (shock mounts removed or blocked) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

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(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.14 Test for equipment vibration.- The equipment shall be tested as follows, to determine conformance to 3.19.6.2:

(a) Secure the equipment on its shock mounts to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position with connections (such as cable assemblies) in place.

(b) Vibrate the equipment successively, in three mutually perpendicular directions, over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be 0.030 inch; however, this may be reduced, if necessary, to prevent excessive excursion at or near resonance provided that the excursion can still be measured.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure amplitude of the applied vibration and the vibration of the equipment, for frequencies between 10 and 25 and between 35 and 55 cycles per second.

(e) Determine the natural frequency of the equipment; the frequency at which the ratio of the equipment vibration to the applied vibration is greatest shall be considered the natural frequency.

4.15 Bounce test; vehicular mounted.- The equipment shall be tested on the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

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(a) Secure the equipment on its shock mount to the 48- by 48-inch vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of 3 hours at 283 ± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction. Rest periods may be provided to permit cooling of the resilient mounts, provided that the testing time is not less than 3 hours.

4.16 Bounce test; field transportable.- The equipment in its carrying case (without shock mount) shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

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(b) Place the equipment in its transit case, if any, prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ± 2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.17 Shock test; bench-handling.- The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing.

(a) Tilt up the assembly through an angle of 30 degrees using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.18 Shock test; drop.- The receiver, in its canvas carrying case, shall be subjected to the drop test in accordance with Table VI. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame.

TABLE VI - DROP TEST

Largest dimension of case (inches)	Notes	Height of drop (inches)	Number of drops
Under 36	A	48	Drop on each face, edge, and corner. Total of 26 drops.
36 and Over	B	30	

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Note A: Drops shall be made from a quick-release hook; or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal.

Note B: Drops shall be made from a quick-release hook.

4.19 Orientation test.- The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 $\pm^{3}_{1}$ degrees. During inclinations in each plane, the equipment shall meet specified performance for the following measurements.

Paragraph	Measurement
4.7.10	AM sensitivity (vehicular operation)
4.7.15	Calibration accuracy
4.7.31	Diode outputs

4.20 Test for shrinkage.- The canvas items shall be tested for conformance to 3.4.2.1.

4.21 Battery compartment watertight test.- Fill the dry battery compartment with tap water. Lock the battery cover plate in place. Set the receiver dust cover on each of its 6 sides; that is: top, bottom, front, rear, right side and left side; observe inside and outside of dust cover for water leakage. (See 3.17)

4.22 Operational inspection.- Immediately prior to packaging for shipment receiver R-902( )/GR shall be operated and receive intelligible signals at one frequency on each band. (See 3.24)

4.23 Inspection for dimensional interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.21. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

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(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.24 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.25 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 tc determine conformance to the requirements of section 5.

4.26 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.1(e)(2)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

4.7.36 Calibration adjustment
4.22 Operational inspection

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C, as specified. (See 6.1(c))

5.1.1 Level A.-

5.1.1.1 Cleaning.- Cleaning shall be accomplished in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying.- Drying shall be accomplished in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Unit packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116.

5.1.1.3.1 Running spares.- Running spares and accessories items as required, shall be packaged as follows:

5.1.1.3.1.1 Electron tubes.- All tubes not securely clamped in place shall be removed and packaged individually in accordance with MIL-E-75. Spare tubes shall be individually packaged in accordance with MIL-E-75.

5.1.1.3.1.2 All other items.- All other items of spare parts shall be packaged in accordance with the requirements of MIL-E-17555.

5.1.1.3.1.3 Consolidated package.- Consolidate the items packaged as specified in 5.1.1.3.1.1 and 5.1.1.3.1.2, within a close-fitting corrugated fiberboard box conforming to PPP-B-636. Fill all voids with paperboard conforming to PPP-P-291, type II, to prevent movement. Box closure shall be as specified in the appendix of the box specification.

5.1.1.3.2 Receiver, Radio R-1419( )/GR.- Each receiver shall be packaged Method II b, as follows; Cushion each receiver on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, grade W5c, designed to absorb the shock of impact in handling and transit. Place the cushioned receiver within a close-fitting fiberboard box conforming to PPP-B-636, domestic class. Distribute the required amount of desiccant, conforming to MIL-P-116, within the cells and pads around the receiver. Seal the box closure with gummed paper tape conforming to UU-T-111. Blunt all corners of the box. Place the sealed interior box within a bag fabricated of water vapor-proof barrier material conforming to MIL-B-131, for class I application. The barrier bag shall have one dimension sufficient to allow three resealings for inspection purposes. A humidity indicator card conforming to MIL-I-8835, shall be placed on top of the interior box and secured with pressure-sensitive tape, conforming to PPP-T-76. Heat seal the barrier bag leaving a small opening to exhaust the air. Exhaust the air and complete the heat seal. Place the water vapor-proof receiver within a close-fitting RS style fiberboard box, Grade 5, conforming to PPP-B-636. Any stapling or stitching on the inside of the exterior box shall be covered with pressure sensitive tape. Seal the box closure in accordance with the appendix of the box specification.

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5.1.2 Level C.- Receiver R-1419( )/GR shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be Level A, B or C, as specified. (See 6.1(c))

5.2.1 Level A.- Receiver, R-1419( )/GR, packaged as specified in 5.1 shall be packed within fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirement use or MIL-B-43291. Waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape; not less than two inches wide, conforming to PPP-T-76. The tape shall be centered over the seams and joints and shall extend over the corners and edges of the box a minimum of two inches onto the adjacent box panels. Tape shall be applied over the lengthwise seams of the outer flaps closing the openings of the box and over the manufacturer's joint prior to tape being applied to the edge seams of the box. The tape applied to the manufacturer's joint shall only fully cover the joint and not be extended over the corners of the box onto the adjacent panels. The shipping containers shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97 in accordance with the appendix of the applicable box specification.

5.2.2 Level B.- Receiver R-1419( )/GR, packaged as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-636, Type CF, class domestic, special requirements use. Closure shall be in accordance with the appendix of the box specification.

5.2.3 Level C.- Receiver R-1419( )/GR, packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. (See 6.1(f))

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging required for shipment. (Level A, Level B, or Level C.) (See section 5)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Three sample units of each item cited in section 1 are generally required so that lengthy environmental and reliability tests can be completed on two sample units while complete performance measurements can be made on the third sample unit. (See 3.3)

(2) Preproduction pack(s).

Make-up of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (See 3.26)

(i) Quantity of tools and running spare parts required. (See 3.26)

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6.2 Color.- The color chip furnished by the procuring agency (3.5) will be color chip No. X-24087, and may be obtained upon request to Commanding Officer, U. S. Naval Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.3 Definition of Level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.5 Location of operational inspection.- It is desirable that the operational inspection (4.22) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: R-1419W/GR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.7 Color of textiles.- Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.8 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provision established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN
Army (EL)

PREPARING ACTIVITY
Army (EL)

PROJECT NUMBER 5820-A337

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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